TACO-Gold: From Trump Retreat Risk to Gold Forecasting

Group 5

ECO6130 Final Project

Repository: github.com/balabababa-nana/taco_gold

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1 Introduction

Political communication can affect market expectations, especially when it concerns trade policy, geopolitical pressure, inflation, or other sources of uncertainty. This report studies whether a narrow pattern in Trump-related policy communication can be converted into a useful forecasting signal for gold returns. The object is not broad real-world policy reversal behavior. It is a *48-hour text-observed retreat* signal: a suspected policy-pressure event followed within 48 hours by clear text-visible softening, delay, exemption, or retreat.

The design is two-stage. Stage 1 converts candidate pressure events into a calibrated event-level probability, P_{taco}. Stage 2 aggregates this probability into daily forecasting inputs and combines it with market controls to predict gold returns. This structure treats Stage 1 as supervised signal compression: noisy political text is first transformed into a mechanism-specific probability before being used in the noisier return prediction task.

The report asks two questions: whether this text-observed retreat pattern is forecasting-relevant for gold returns, and whether the two-stage design outperforms direct single-stage text-to-return and macro-only alternatives. Under the corrected rerun design, the selected two-stage model is slightly ahead of the designated holdout forecasting benchmarks, and the retained dense-export robustness check does not overturn this conclusion. A stylized strategy test is used only as a supplementary economic-value check, where the two-stage rule is strongest at a 2.5 bps forecast threshold but not uniformly dominant across thresholds. The central claim is therefore modest: a narrow political text pattern can be operationalized as a calibrated intermediate signal, and in this setting that compressed signal is useful for gold return forecasting.

2 Literature

This project builds on the finance and economics literature that treats text as structured data rather than as qualitative background. Prior work provides the broad methodological foundation for transforming high-dimensional language into measurable economic variables and places this approach within the modern finance NLP literature (Gentzkow, Kelly, and Taddy, 2019; Hoberg and Manela, 2025). In asset pricing, media language and news-based uncertainty measures have also been shown to contain market-relevant information (Tetlock, 2007; Manela and Moreira, 2017). These studies justify the basic premise that political and news text can contain information relevant for financial markets.

A closer strand constructs interpretable text-based policy or political-risk measures before studying downstream outcomes. Economic policy uncertainty and firm-level political risk are both examples of text being compressed into economically meaningful intermediate variables rather than used only as raw language (Baker, Bloom, and Davis, 2016; Hassan et al., 2019). At the same time, another strand maps text more directly into returns. Supervised text-to-return prediction is feasible at large scale, but direct supervision on noisy return targets can be sensitive to topic design and overfitting (Ke, Kelly, and Xiu, 2019; Glasserman, Krstovski, and Laliberte, 2020). The implication for this project is not that two-stage designs are generally superior. Rather, the literature motivates a compressed-signal design as a reasonable alternative whose advantage must be tested empirically.

The first stage of this project also connects to rare-event classification and probability evaluation. The positive label is sparse because clear short-window retreat evidence is uncommon. Rare-event settings require care, and standard classification framing can be misleading (King and Zeng, 2001). Precision-recall evaluation is also more informative than ROC-based evaluation when the positive class is rare (Saito and Rehmsmeier, 2015). This supports the use of PR-AUC in Stage 1. Because Stage 2 consumes P_{taco} as a continuous probability signal, probability quality also matters; this motivates reporting calibration-oriented metrics such as Brier score and expected calibration error (ECE) rather than relying only on hard-label accuracy.

The gap addressed here is therefore narrow. Existing work shows that text matters for markets, that political and policy text can be transformed into structured risk measures, and that direct text-to-return prediction is possible. This report combines those ideas around a specific political mechanism: a 48-hour text-observed retreat pattern. Its contribution is to operationalize that mechanism as a calibrated event-level probability, aggregate it into a daily signal, and test whether the resulting compressed text signal improves gold return forecasting relative to both direct text and no-text alternatives.

3 Data, Label, and Research Design

The project combines Trump public text, gold prices, market stress controls, and attention proxies. Table 1 summarizes the main data blocks. The text data provide the event anchors and follow-up windows used to define the retreat signal, while the market data provide the daily target and control variables used in the forecasting stage.

Table 1: Data Sources

Data block	What we collect	Primary source
Trump first-term text	Tweets, timestamps, raw text	Kaggle dataset
Trump second-term text	Truth Social posts, timestamps, raw text	Public GitHub archive
Market stress	SPX drawdown, VIX change, related controls	FRED
Gold target series	Gold daily close and return series	Alpha Vantage
Attention proxy	Tariff / war / inflation search attention	Google Trends

The key label is event-level rather than post-level. A candidate event begins with a suspected policy-pressure text, such as a tariff threat, geopolitical warning, or other pressure-oriented statement. The labeling window then looks at Trump's subsequent public text over the next 48 hours. A positive `label_retreat` is assigned only when the follow-up text contains clear evidence of softening, delay, exemption, narrowed scope, cancellation, or retreat. Generic optimism, vague negotiation language, or relationship improvement without a clear policy action is not sufficient.

This definition is deliberately text-observed. A negative label does not prove that no real-world retreat occurred; it only means that the observed 48-hour text window does not contain sufficiently clear retreat evidence under the project definition. This distinction is important because the project is designed to build a market-alignable text signal, not a complete historical database of policy reversals.

The design uses two units of observation. Stage 1 is an event-level rare-event classification problem: candidate pressure events are converted into calibrated probabilities, P_{taco}, of a 48-hour text-observed retreat. Stage 2 is a daily return-forecasting problem: event-level probabilities are aggregated to the trading-day level and combined with market controls to predict gold returns.

The chronological split is designed to avoid leakage across the platform and administration gap:

- First-term training window: 2017-01-20 to 2021-01-08.
- Excluded gap / Biden period: 2021-01-09 to 2025-01-19.
- Second-term training window: 2025-01-20 to 2025-07-14.
- Primary holdout window: 2025-07-15 to 2025-10-25.

Validation is segmented and gap-aware. Model selection does not cross the excluded Biden-period gap, and Stage 2 uses only legal out-of-fold or true holdout P_{taco} values. This rule is central to the empirical design because the downstream return model must not observe event probabilities generated from future information.

4 Method Overview

The modeling pipeline has three steps. First, the raw Trump text corpus is filtered into high-recall candidate pressure events. Second, each candidate event is labeled using the anchor text and its 48-hour follow-up window, producing the Stage 1 target. Third, the fitted Stage 1 model exports legal event-level probabilities, which are aggregated into daily inputs for the Stage 2 gold return model.

Single-stage design:

raw / daily text representation → gold return prediction

Two-stage design:

text + pressure features → Stage 1 P_{taco} → daily aggregation → Stage 2 gold return prediction

Figure 1: One-Stage vs Two-Stage Design. Stage 1 acts as a supervised signal-compression layer that converts event-level political text into a calibrated retreat probability before the daily return forecasting task.

Figure 1 highlights the main structural choice. The single-stage alternative maps text-derived daily features directly into returns. The proposed two-stage design instead uses Stage 1 as a supervised signal-compression layer, converting event-level political text into a calibrated retreat probability before the daily return forecasting task.

Stage 1 is treated as a rare-event probability problem. Because positive retreat events are sparse, standard accuracy is not the main criterion. The model selection rule jointly considers ranking quality and probability quality, with emphasis on PR-AUC, Brier score, and calibration error. This is important because the downstream model uses P_{taco} as a continuous signal rather than only as a hard event classification.

Stage 2 is a daily supervised forecasting problem. Its target is the gold daily return, and its inputs combine daily aggregates of legal P_{taco} with lagged market and macro-financial controls. The main forecasting comparison is therefore not only a model comparison, but also a design comparison: the proposed pipeline is tested against direct text-to-return prediction and against no-text macro-only benchmarks. Table 2 summarizes the main input blocks by stage.

Table 2: Model Inputs by Stage

Stage	**Input block**	**Role in the model**
Stage 1	Anchor-text sentiment and pressure features	Describe the suspected pressure event
Stage 1	Candidate/event metadata	Locate and characterize the event window
Stage 1	Lagged pressure and attention proxies	Capture recent policy-pressure context
Stage 2	Daily P_{taco} aggregates	Carry the compressed text-derived signal
Stage 2	Lagged market and macro controls	Capture non-text financial conditions
Stage 2	Cross-asset structure	Control for broader market dynamics

All Stage 2 comparisons use the same leakage discipline: training-side daily P_{taco} values must come from out-of-fold Stage 1 predictions, while holdout values must come from models trained without holdout labels. This keeps the two-stage signal comparable to the single-stage and macro-only alternatives.

5 Main Results

This section asks whether the text-defined retreat pattern can be turned into a forecasting-relevant signal for gold returns. The evidence is presented in two steps: first, whether Stage 1 produces a usable calibrated probability signal; second, whether the selected Stage 2 model improves holdout return forecasts relative to the designated benchmarks.

Stage 1 selects linear SVM with sigmoid calibration. The selection is based on a joint reading of PR-AUC, Brier score, and ECE, rather than any single metric alone. This is appropriate because the first-stage task is imbalanced and because the downstream return model consumes probabilities, not only hard classifications. Table 3 reports the full calibrated-variant ranking.

Table 3: Stage 1 Model Selection

Rank	**Model**	**Prob.**	**PR-AUC**	**Brier**	**ECE**
1	linear_svm	sigmoid	0.119414	0.056294	0.056772
2	logistic_regression	sigmoid	0.107069	0.057551	0.058287
3	linear_svm	isotonic	0.105717	0.058101	0.061506
4	logistic_regression	isotonic	0.091311	0.060181	0.074243
5	xgboost_classifier	isotonic	0.083942	0.053225	0.038564
6	xgboost_classifier	sigmoid	0.083826	0.053299	0.039111

The selected Stage 1 variant remains usable on the primary holdout, with PR-AUC = 0.153295, Brier = 0.045794, and ECE = 0.018409. The holdout PR-AUC is above the outer-CV reading, and the low holdout ECE suggests that the probability signal does not suffer an obvious calibration collapse in the new window. I interpret this narrowly: the classifier produces a non-trivial event-level probability signal for downstream aggregation, not a standalone political-behavior model.

Stage 2 uses the selected `linear_svm__sigmoid` signal and chooses `random_forest + two_stage_v2_core` as the final forecasting model. In outer validation, the selected model has RMSE = 0.009845, which provides the basis for carrying it into the frozen holdout comparison. The main report-facing result is the primary holdout comparison in Table 4.

Table 4: Main Stage 2 Forecasting Results

Model	**RMSE**	**MAE**	**OOS R^2**	**Sign Acc.**
two-stage final	**0.014453**	0.009367	**0.007664**	**0.648649**
RandomForest + macro_only	0.014471	**0.009331**	0.005295	0.635135
Ridge + macro_only	0.014610	0.009379	-0.013962	0.581081
Random Walk (Zero)	0.014606	0.009495	-0.013407	0.000000

On the primary holdout, the selected two-stage model has the lowest RMSE among the designated main benchmarks. The margin over the strongest macro-only benchmark is small, but it is directionally consistent: 0.014453 for the two-stage final versus 0.014471 for RandomForest + macro_only. The two-stage model also remains ahead of the ridge macro-only model and the random-walk-zero benchmark on RMSE.

These results answer the first research question in a limited but positive way. The narrow 48-hour text-observed retreat pattern can be transformed into a calibrated event-level signal, and that signal is useful enough to improve the main holdout return forecast relative to the benchmark set used in the corrected rerun design. The next section tests whether this improvement is specifically tied to the two-stage structure rather than to text information or macro controls alone.

6 Core Structural Comparison and Dense-Export Robustness

The central design question is whether the intermediate P_{taco} layer adds value relative to simpler alternatives. I use two challengers for this purpose. The single-stage model tests whether a direct daily text representation can match the proposed compressed-signal design. The macro-only benchmark tests whether the text-derived signal adds information beyond standard market controls.

The formal single-stage comparator is `RandomForest + single_stage_daily_raw_rich`. It uses a rerun-aligned direct text-to-return setup rather than the event-level P_{taco} layer. Table 5 compares this model with the selected two-stage final and the strongest macro-only benchmark on the same primary holdout.

Table 5: Structural Comparison and Dense-Export Robustness

Model	**Role**	**RMSE**	**MAE**	**OOS R^2**	**Sign Acc.**
two-stage final	main model	**0.014453**	0.009367	**0.007664**	**0.648649**
formal single-stage RF	direct text alternative	0.014530	0.009345	-0.002956	0.621622
RandomForest + macro_only	no-text benchmark	0.014471	**0.009331**	0.005295	0.635135
dense-export two-stage	robustness variant	0.014462	0.009369	0.006439	0.581081

The selected two-stage model is slightly ahead of the formal single-stage comparator on holdout RMSE: 0.014453 versus 0.014530. It also has higher sign accuracy. This supports the idea that the supervised intermediate probability is doing useful compression, rather than merely adding complexity. The comparison with RandomForest + macro_only is narrower, but the two-stage model remains slightly ahead on RMSE and OOS R^2. This suggests that the text-derived signal contributes information not fully captured by the macro-control set.

The dense-export robustness check tests whether this conclusion depends on sparse legal train-side P_{taco} coverage. The robustness variant uses denser forward-only Stage 1 validation blocks to create more legal out-of-fold probabilities without relaxing the no-leakage rule. In other words, it changes how many legal train-side probability exports are produced, not the holdout labels or the future-information boundary. This increases event-level OOF coverage from 36.7% to 78.8%, positive-event OOF coverage from 56.4% to 90.9%, and positive target-day coverage from 14/22 to 18/22. The dense-export two-stage model remains close to the mainline result and does not overturn the comparison against the single-stage and macro-only alternatives.

The interpretation should therefore be structural but modest. The evidence is consistent with the value of supervised intermediate signal compression in this setting. It does not show that two-stage architectures are universally superior, and the margins are small. The stronger conclusion is narrower: under the corrected rerun design, the direct text alternative and the no-text benchmark do not overturn the selected two-stage model.

7 Economic Value Check

I complement the forecasting results with a stylized economic-value check. The exercise is a simple long/flat rule, not a fully optimized trading system. Using the primary holdout predictions, the strategy takes a long position when the model forecast exceeds a fixed forecast threshold and otherwise stays flat. Table 6 reports two thresholds, 2.5 bps and 5 bps, both with 5 bps transaction costs.

Table 6: Stylized Strategy Check

Threshold	Strategy	Net Cum. Return	Net Sharpe	Max Drawdown	Turnover	Entries
2.5 bps	two-stage final	**30.87%**	**5.713**	-3.77%	16	8
2.5 bps	dense-export two-stage	22.65%	4.576	-4.12%	26	13
2.5 bps	single-stage RF	19.37%	2.686	-8.06%	9	5
2.5 bps	macro_only RF	24.03%	3.248	-8.02%	7	4
2.5 bps	random_walk_zero	0.00%	NA	0.00%	0	0
5 bps	two-stage final	16.26%	4.017	-2.99%	42	21
5 bps	dense-export two-stage	20.22%	5.585	-2.12%	32	16
5 bps	single-stage RF	13.37%	2.031	-10.91%	33	17
5 bps	macro_only RF	**31.25%**	**7.382**	-1.52%	18	9
5 bps	random_walk_zero	0.00%	NA	0.00%	0	0

The two-stage final produces positive net performance and remains ahead of the single-stage RF strategy in this simplified setting. At the 2.5 bps forecast threshold, the mainline two-stage strategy is also the strongest strategy in the table by net cumulative return and Sharpe. At the stricter 5 bps threshold, however, the macro-only RF strategy is stronger. Therefore, the economic-value check should be read as limited actionability evidence and as threshold-sensitive support, not as proof that the two-stage model is uniformly the best trading rule.

8 Conclusion

This report studies whether a narrow 48-hour text-observed retreat pattern in Trump-related policy communication can be converted into a useful forecasting signal for gold returns. The evidence supports a cautious positive answer. Stage 1 turns candidate pressure events into a calibrated event-level P_{taco} signal, and Stage 2 uses that signal to improve the primary holdout return forecast relative to the designated benchmark set.

The structural comparison is the main contribution. The selected two-stage model remains slightly ahead of both the formal single-stage text-to-return alternative and the strongest macro-only forecasting benchmark under the corrected rerun design. The dense-export robustness check does not overturn this conclusion. The stylized strategy test further suggests that the signal has directional value: the two-stage rule is strongest at the 2.5 bps forecast threshold, while macro-only RF is stronger at the stricter 5 bps threshold. This makes the economic-value evidence supportive but threshold-sensitive.

The findings are limited by the narrow label definition, the short second-term holdout, and the fact that the target is text-observed retreat rather than verified real-world policy reversal. Within those limits, the project shows that a mechanism-specific compressed text signal can add useful structure to gold return forecasting.

Appendix

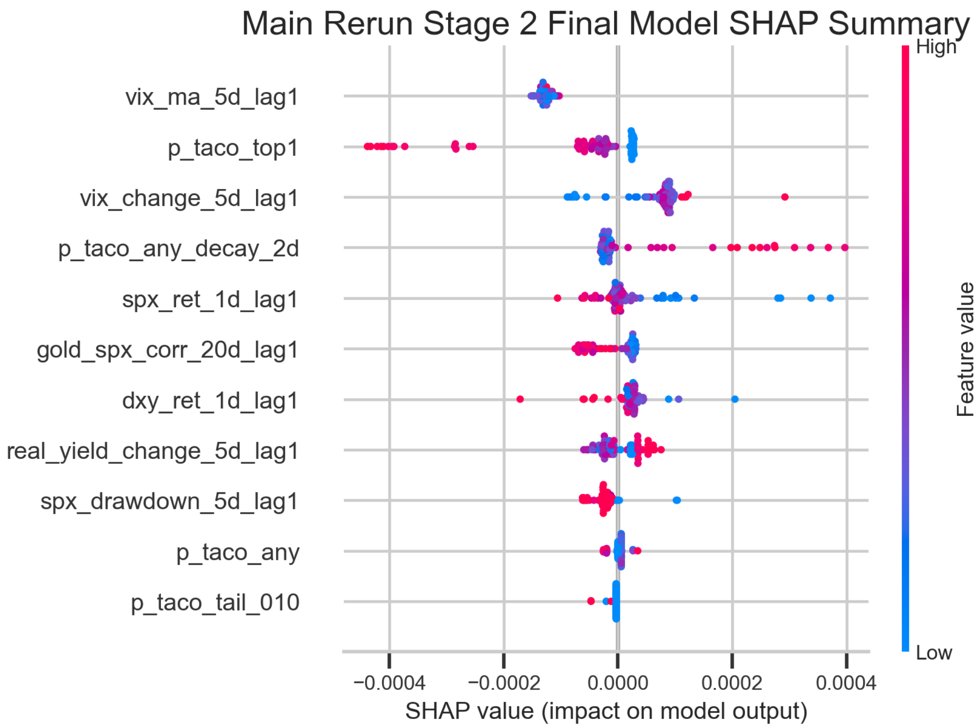

Figure A1: SHAP summary for the selected Stage 2 two-stage model on the holdout sample.

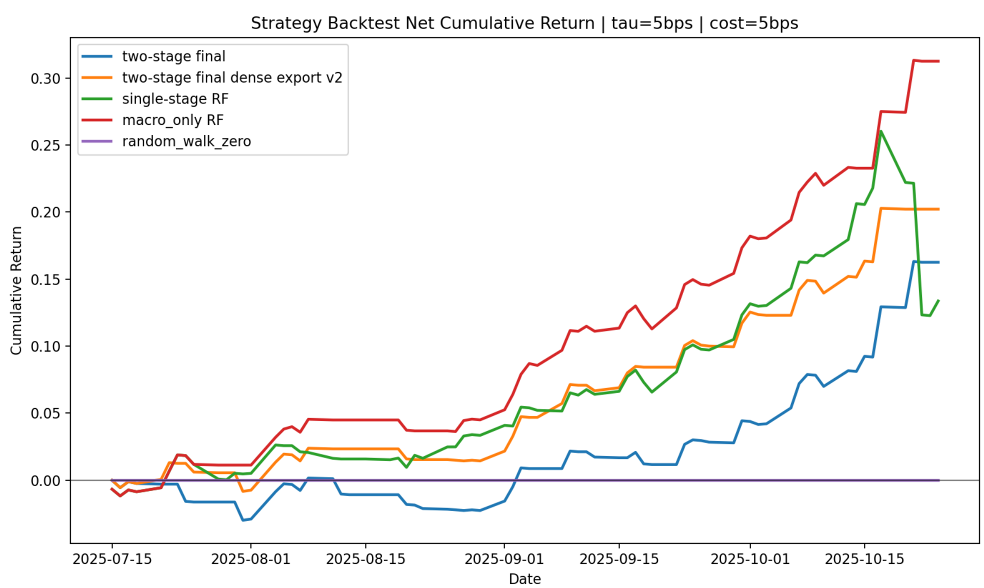

Figure A2: Net cumulative return for the stylized strategy comparison using a 5 bps forecast threshold and 5 bps transaction costs.

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